



StageAgent

Personalized Theater Recommendation & Planning Agent

MATH 690 - Graduate Seminar : Agentic AI Project

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Agenda

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Problem Statement

Current Challenges

Theater enthusiasts face multiple pain points:

-  Hard to track which plays are showing when
-  Difficult to filter by location/distance
-  No personalized recommendations based on past preferences
-  Missing context (actor interviews, reviews)
-  Manual calendar coordination

Our Solution

An AI agent that:

- Syncs with your calendar
- Recommends plays based on your taste
- Filters by location and availability
- Suggests pre/post-show content (interviews, reviews)



Solution Overview

Core Features

1. Calendar Integration

- Automatically finds free time slots
- Avoids scheduling conflicts

2. Intelligent Filtering

- Location-based (max 30 min travel)
- Genre preferences (drama vs. comedy)
- Venue preferences (state theater vs. private)

3. Personalized Scoring

- Analyzes past ratings
- Predicts enjoyment likelihood

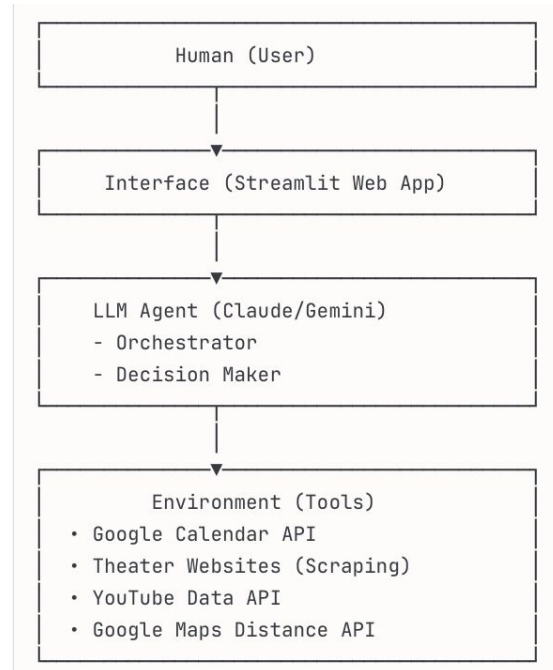
4. Content Discovery

- Actor interviews (YouTube)
- Reviews and articles
- Behind-the-scenes content



Technical Architecture

High-Level Design (From GSU - Agents 2 Lecture Slide 16)





Prompting Techniques Used

1. Zero-Shot (from Slide 6)

Use Case: Simple queries

Examples: "What plays are showing today in Istanbul?"

2. Few-Shot (from Slide 7)

Use Case: Preference prediction

Examples:

- "Hamlet" → Liked (Drama, Classic)

- "Comedy Night" → Disliked (Shallow humor)

Predict: Will user like "King Lear"?

3. Chain-of-Thought (from Slide 8)

Use Case: Multi-step reasoning

1. Analyze user's calendar 2. Find available time slots 3. Filter plays by location 4. Score based on past preferences 5. Rank and recommend



System Components

Tool Calling (Slide 12)

Possible Available Tools:

```
# Calendar Management
get_calendar_events(date_range)

# Theater Data
search_plays(city, date)
get_play_details(play_id)

# Location Services
calculate_distance(user_location, venue)

# Content Discovery
search_interviews(play_name, actor)
get_reviews(play_name)

# User Preferences
get_user_ratings(user_id)
save_user_rating(play_id, rating)
```



System Components



Agentic Architecture Pattern

Planner Pattern (Slide 23)

User Query: "Plan my theater schedule for December"



[Planner LLM]



[Synthesizer]



Monthly Plan with 4-5 play recommendations

****Why Planner?**

- Complex task requiring multiple steps
- Tasks can be executed **in** sequence
- Plan needs to be tracked across iterations



System Components



Reflection Loop (Slide 21)

Continuous Improvement

User: "Recommend a play"



[Generator]: "I suggest 'Hamlet'"



[Reflector]: "Check if user likes tragedy"



User Feedback: ★★★★★



[Update Model]: Increase weight for tragedy genre

**Benefits:

- Learns from user feedback
- Improves over time
- Self-correcting system

System Components

Technical Stack

i) Backend

- Language: Python 3.11+
- LLM Integration: LiteLLM (multi-provider)
- Models: Claude 3.5 Haiku, Gemini 2.0 Flash
- APIs: - Google Calendar API - YouTube Data API v3 - Google Maps Distance Matrix API

ii) Frontend

- Framework: Streamlit
- Deployment: Streamlit Cloud

iii) Data Storage

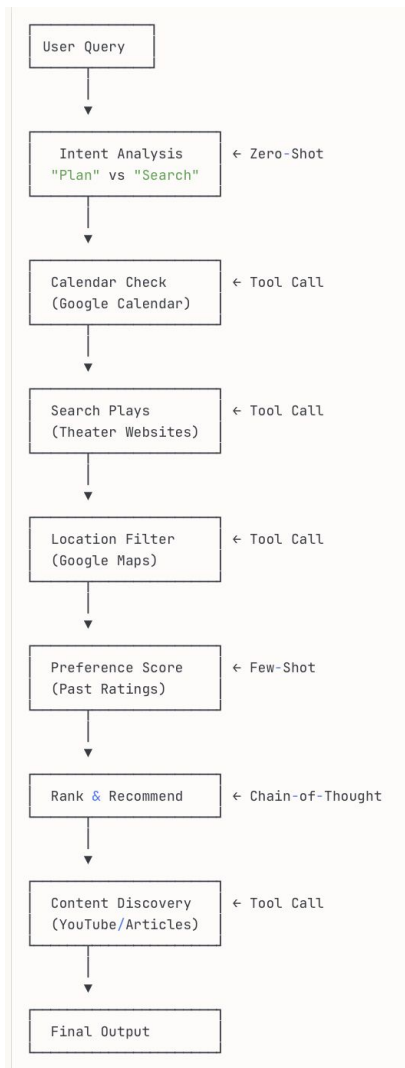
- User Preferences: SQLite
- Cache: Redis (for API responses)



System Components



System Flow Diagram



Theater Recommendation System - Project Timeline

Week	Task	Start	End
WEEK 1	Project Kickoff & Research	10/10/2025	16/10/2025
	Define project scope	10/10/2025	12/10/2025
	Research existing systems	10/10/2025	14/10/2025
	Development environment setup	13/10/2025	16/10/2025
	Prepare project proposal	14/10/2025	16/10/2025
	Friday presentation prep	16/10/2025	16/10/2025
WEEK 2	Data Collection & Basic Implementation	17/10/2025	23/10/2025
	Web scraping development	17/10/2025	20/10/2025
	Google Calendar API integration	17/10/2025	21/10/2025
	Database schema creation	19/10/2025	22/10/2025
	Zero-shot prompting implementation	20/10/2025	23/10/2025
	Friday presentation prep	23/10/2025	23/10/2025
WEEK 3	Core Recommendation Engine	24/10/2025	30/10/2025
	Few-shot prompting development	24/10/2025	27/10/2025
	Google Maps API integration	25/10/2025	28/10/2025
	Scoring algorithm	27/10/2025	29/10/2025
	Testing with 10 sample plays	28/10/2025	30/10/2025
	Friday presentation prep	30/10/2025	30/10/2025
WEEK 4	Agentic Patterns Implementation	31/10/2025	06/11/2025
	Tool calling loop development	31/10/2025	03/11/2025
	Planner architecture	01/11/2025	04/11/2025
	Reflection loop integration	03/11/2025	05/11/2025
	Integrate all components	04/11/2025	06/11/2025
	Friday presentation prep	06/11/2025	06/11/2025
WEEK 5	Chain-of-Thought & Content Discovery	07/11/2025	13/11/2025
	CoT prompting implementation	07/11/2025	10/11/2025
	YouTube API integration	08/11/2025	11/11/2025
	UI development (Streamlit)	09/11/2025	12/11/2025
	Feedback mechanism	11/11/2025	13/11/2025
	Friday presentation prep	13/11/2025	13/11/2025
WEEK 6	Evaluation & Testing	14/11/2025	20/11/2025
	User testing (5 participants)	14/11/2025	18/11/2025
	Metrics collection and analysis	15/11/2025	19/11/2025
	Prompting techniques comparison	16/11/2025	19/11/2025
	Performance benchmarking	17/11/2025	20/11/2025
	Friday presentation prep	20/11/2025	20/11/2025
WEEK 7	Final Integration & Documentation	21/11/2025	27/12/2025
	Travel module integration	21/11/2025	28/11/2025
	Documentation writing	24/11/2025	10/12/2025
	Final presentation preparation	05/12/2025	20/12/2025
	Deploy to Streamlit Cloud	10/12/2025	15/12/2025
	Final report writing	15/12/2025	26/12/2025
	FINAL PRESENTATION & SUBMISSION	27/12/2025	27/12/2025

Risk Management

1. Technical Risks		
(Risk)	(Impact)	(Probability)
API Rate Limits (Google Calendar, Maps)	High	Medium
LLM Hallucinations (Incorrect recommendations)	Medium	Medium
Model Cost Overrun (Claude API expensive)	Medium	Low

2. Data Quality Risks		
Outdated Play Information	High	High
Incomplete Theater Database	Medium	Medium
Web Scraping Failures (Site structure changes)	Medium	Medium

3. Integration Risks		
Calendar API Downtime	Medium	Low
Maps API Changes/Costs	Medium	Medium
YouTube API Quota Exceeded	Low	Low

4. User Experience Risks		
Poor Recommendation Quality (New users with no history)	Medium	High
Privacy Concerns (Calendar access)	High	Medium
Slow Response Time (>10 sec)	Medium	Medium

5. Project Timeline Risks		
Week 3-4 Delays (API integration complex)	High	Medium
Final Week Crunch (Integration issues)	Medium	Medium

6. Budget & Cost Risks		
API Costs Exceed Budget	Medium	Low
Unexpected API Charges	Low	Low



Evaluation Metrics

Accuracy

- Recommendation Hit Rate: (% of recommended plays actually attended)
- Target: (i.e. 70% and above)

User Satisfaction

- Average Rating: (User feedback on recommendations)
- Target: (i.e. 4.2/5.0)

System Performance

- Response Time: (Time to generate plan)
- Target: (i.e. < 10 seconds)

Coverage

- Play Database: Number of plays tracked
- Target: (i.e. 50 plus active plays)

(Expected) Learning Outcomes

1. Agentic AI Patterns

- Planner architecture
- Tool calling integration
- Reflection loops

2. Prompting Techniques

- Zero-shot **for** quick queries
- Few-shot **for** preference learning
- Chain-of-thought **for complex** reasoning

3. Real-World Integration

- API integration (Google Calendar, Maps)
- Web scraping (theater sites)
- Data persistence (user preferences)



Future Work

Phase 1 (Current Project)

-  Single city (Istanbul)
-  Manual calendar sync
-  Basic preferences

Phase 2

-  Multi-city support
-  International festivals
-  Travel cost estimation
-  Hotel/flight booking

Phase 3 (Advanced Features)

-  Group recommendations (friends)
-  Automatic ticket purchasing
-  Mobile app (React Native)
-  Multi-language support

References

Course Materials:

- Lecture Slides: "Prompting LLMs" (Week 3)
- Lecture Slides: "Architecture"(Week 4)

Technical Resources:

- Anthropic Claude Documentation
- LiteLLM Documentation - Google Calendar API v3
- YouTube Data API v3

Inspiration:

- Dr.Erdem Özcan

 Thank You!

Contact

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Will be Available At [theater-planner-demo.streamlit.app]



Possible Common Questions:

Q1: How do you handle multiple users?

A1: Each user has isolated preference profile in database

Q2: What if play info is incorrect?

A2: Reflection loop allows users to report errors, system self-corrects

Q3: Privacy concerns with calendar access?

A3: OAuth2, read-only access, user can revoke anytime

Q4: Cost of API calls?

A4: Cached responses, batched requests, ~\$5/month for 100 users